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Error Detection and Correction Code using Improved Redundant Matrix Code Algorithm

Abstract

➤ Mobile way communication over the Internet requires error detection and correction. Error Control describes how networks handle and detect errors at the data link layer .

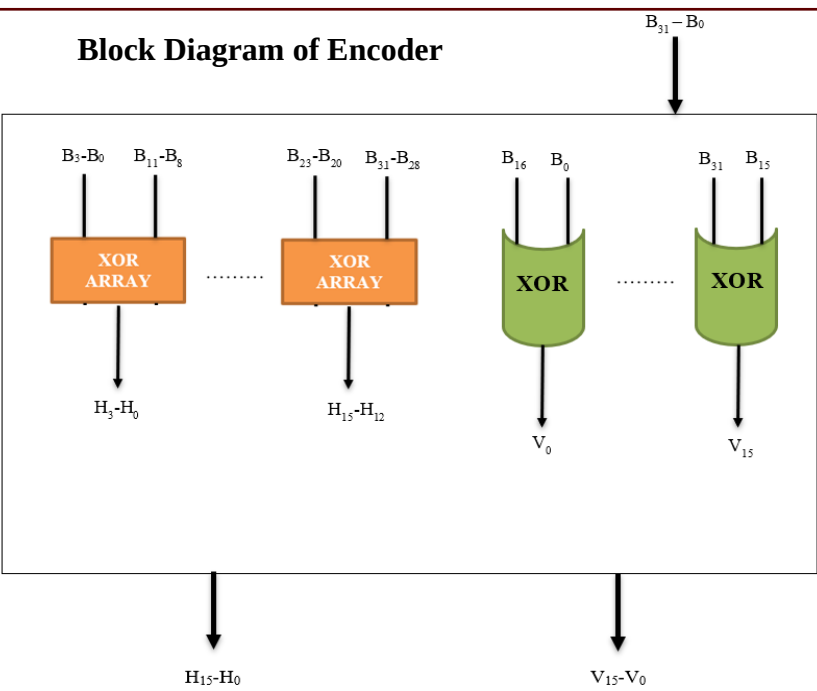
➤ Each network connection is assumed to have two channels. One for data traffic and one for signaling and control. Robust error detection is encoded on the traffic channel based on the information bits and the correction code to form the transmitted data stream.

➤ The proposed technique is called Improved Redundant Matrix Code to protect SRAM-based memories from the detuning of a large number of components.

➤ The Improved Redundancy Matrix Code (IRMC) is used to detect the and troubleshoot storage devices. IRMC's detection rate and adjustments are in contrast to other rescue codes.

➤ The proposed IRMC technique is performed using Xilinx tool which consists of Verilog HDL language.

Block Diagram of Encoder

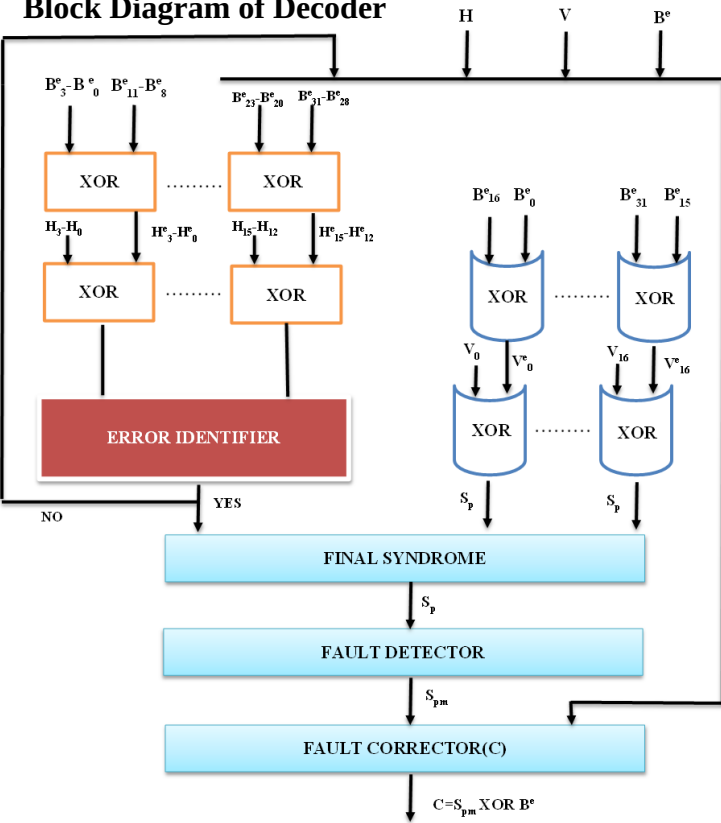


The above mentioned figure is block diagram of encoder of IRMC algorithm. It is used to generate horizontal and vertical redundant bits, which will be very helpful in decreasing redundancy while decoding the erroneous code.

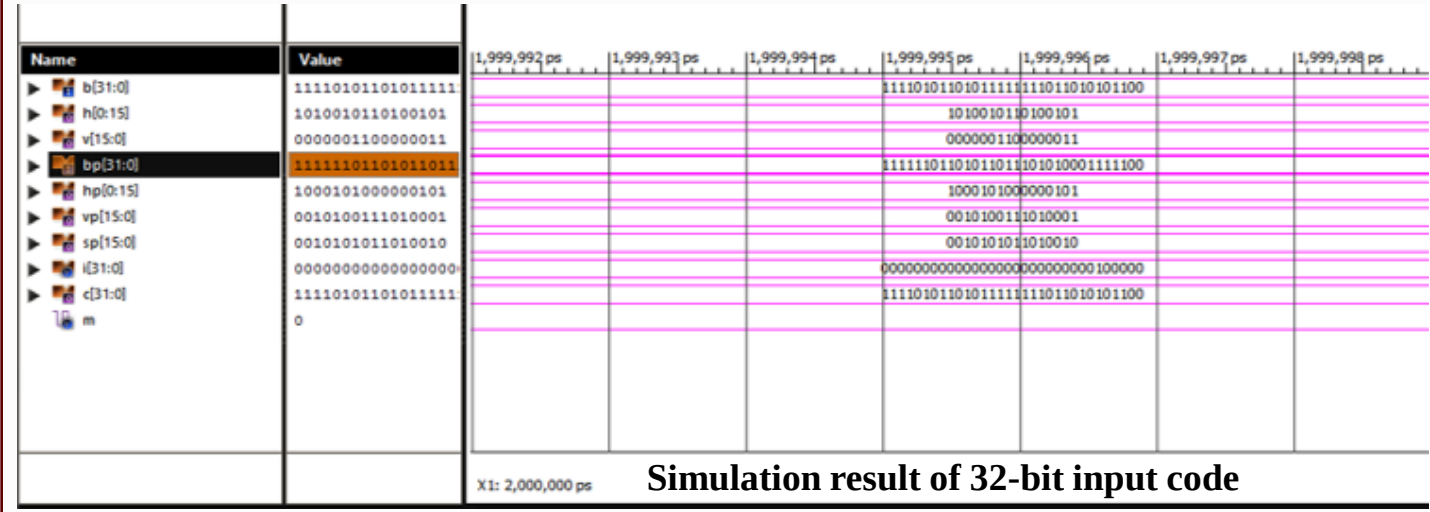
Methodology

1. An original input (B) code (32-bit) is considered. It is then encoded to generate redundant bits (16-bits).
2. An erroneous input is considered along with original input redundant bits as inputs for decoder.
3. The decoder region generates the redundant bits for erroneous input and analyses whether error using error identifier.
4. If error exists, using the syndrome value, error gets detected and corrected.
5. The result (C) of decoder should be equal to B.

Block Diagram of Decoder



Results



Simulation result of 32-bit input code

Error bit correction															
	31 st bit	15 th bit												0 th bit	16 th bit
B	1	1	1	1	0	1	1	0	1	0	1	0	1	1	0
V	1	1	1	1	0	1	0	1	1	0	1	0	1	1	1
B'	0	0	0	0	0	0	1	1	0	0	0	0	0	0	1
B''	1	1	0	1	0	1	0	0	0	1	1	1	1	1	0
V'	1	1	1	1	1	0	1	1	0	1	0	1	1	0	1
S _p	0	0	1	0	1	0	1	0	1	1	0	1	0	0	1
C	0	0	1	0	1	0	1	0	1	1	0	1	0	0	1
	1	1	1	1	0	1	1	0	1	0	1	0	1	1	0
	1	1	1	1	0	1	0	1	1	0	1	0	1	1	1

Logical Organization																							
-----Original bit (32-bit)-----								Horizontal Redundant (16-bit)															
B ₁₅ B ₁₄ B ₁₃ B ₁₂ (S ₃)				B ₁₁ B ₁₀ B ₉ B ₈ (S ₂)				B ₇ B ₆ B ₅ B ₄ (S ₁)				B ₃ B ₂ B ₁ B ₀ (S ₀)				H ₇ H ₆ H ₅ H ₄				H ₃ H ₂ H ₁ H ₀			
B ₃₁ B ₃₀ B ₂₉ B ₂₈ (S ₇)				B ₂₇ B ₂₆ B ₂₅ B ₂₄ (S ₆)				B ₂₃ B ₂₂ B ₂₁ B ₂₀ (S ₅)				B ₁₉ B ₁₈ B ₁₇ B ₁₆ (S ₄)				H ₁₅ H ₁₄ H ₁₃ H ₁₂				H ₁₁ H ₁₀ H ₉ H ₈			
V ₁₅ V ₁₄ V ₁₃ V ₁₂				V ₁₁ V ₁₀ V ₉ V ₈				V ₇ V ₆ V ₅ V ₄				V ₃ V ₂ V ₁ V ₀				-----				-----			

Delay (for 32-bit input): 5.937ns
Power (for 32-bit input): 38.903mW

Team

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